Exercise 5 :

For an ARMA(p, q), we have $\phi(B) Y_t = \theta(B) e_t$

and $Y_t = \psi(B) e_t$ then $\phi(B)[\psi(B) e_t] = \theta(B) e_t$

so $\phi(B) = \dfrac{\theta(B)}{\psi(B)}$ and $\forall x \in \mathbb{R}$ s.t $|x| < 1$

$$\boxed{\phi(x) = \frac{\theta(x)}{\psi(x)}}$$

similarly $e_t = \pi(B) Y_t \Rightarrow \phi(B) Y_t = \theta(B)[\pi(B) Y_t]$

so $\phi(B) = \theta(B)\pi(B)$

finally $\forall x$ s.t $|x| < 1$, $\boxed{\pi(x) = \dfrac{\phi(x)}{\theta(x)}}$